

CS5275 The Algorithm Designer's Toolkit
S2 AY2025/26
Final Exam

5:00 pm – 7:00 pm (2 hours), April 30 (Thursday)

Instructions:

1. **Do not** turn over any page of this paper until instructed to begin.
2. This is an **open-book** assessment; however, the use of calculators or any electronic devices is **not permitted**.
3. The exam consists of 13 true/false questions, 3 multiple-choice questions, and 3 proof-based questions, for a total of 25 marks (worth 40% of the grade).
4. Please write your answers on the answer sheet.
5. All graphs are assumed to be simple (i.e., without self-loops or multiple edges) and undirected.
6. A Boolean function $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ is **constant** if $(f(x) = 1 \text{ for all } x \in \{-1, 1\}^n) \text{ or } (f(x) = -1 \text{ for all } x \in \{-1, 1\}^n)$.
7. In the definition of mixing time $T_{\text{mix}}(G, \epsilon)$, we consider **lazy random walks**.
8. In True-or-false Questions 12–13, we consider the **specific** multiplicative weights update algorithm for a finite two-player zero-sum game discussed in class.

True or False (13 marks: 1 mark per question)

Which of the following statements are correct?

For each statement, indicate whether it is **true** (T or $\bigcirc$) or **false** (F or $\times$).

1. If a graph G has diameter $\Omega(\sqrt{n})$, then the conductance of G is $O\left(\frac{\log n}{\sqrt{n}}\right)$.
2. $T_{\text{mix}}(G, 0.01) \in O(1)$ implies $\Phi(G) \in \Omega(1)$.
3. Select two vertices u and v independently and uniformly at random (with replacement) from any 6-regular n -vertex Ramanujan graph. Then $\mathbb{E}[\text{dist}(u, v)] \in o(\log n)$.
4. There exists a Boolean function $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ whose distance to every parity function $\{\chi_S : S \subseteq [n]\}$ is exactly $1/2$.
5. There exists a Boolean function $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ such that $\hat{f}(S) \neq 0$ for all $S \subseteq [n]$.
6. Every monotone and odd Boolean function $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ is unanimous.
7. In a weighted majority function

$$f(x) = \text{sgn}(a_0 + a_1x_1 + \cdots + a_nx_n),$$

if $a_i \neq 0$ for every $i \in [n]$, then every voter i has positive influence, i.e., $\text{Inf}_i[f] > 0$.

8. For any Boolean function $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ and any $\epsilon \in (0, 1)$, there exists a collection $\mathcal{F}$ of subsets of $[n]$ with $|\mathcal{F}| \in n^{O(1)} \cdot \epsilon^{-O(1)}$ such that the Fourier spectrum of f is ϵ -concentrated on $\mathcal{F}$.
9. For any Boolean function $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$, if $\text{Inf}_i[f] \leq \epsilon/n$ for every $i \in [n]$, then f is $O(\epsilon)$ -close to a constant function.
10. For any Boolean function $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$, if f is ϵ -close to a constant function, then $\text{Inf}_i[f] \in O(\epsilon/n)$ for every $i \in [n]$.
11. In class, we showed that for any deterministic algorithm for learning from expert advice with $n = 2$ experts, an adversary can force the number of mistakes to satisfy $M \geq 2 \min_{i \in [n]} M_i$. The same lower bound $M \geq 2 \min_{i \in [n]} M_i$ also holds when there are $n = 10$ experts.
12. In the analysis of the multiplicative weights update algorithm for a finite two-player zero-sum game discussed in class, we constructed a near-optimal row mixed strategy and showed that there exists $t \in [T]$ such that

$$\max_j A(\mathbf{p}^{(t)}, j) \leq \lambda^* + \epsilon.$$

Suppose instead that t is chosen uniformly at random from $[T]$. Then

$$\mathbb{E} \left[\max_j A(\mathbf{p}^{(t)}, j) \right] \leq \lambda^* + \epsilon.$$

13. In the same analysis, we constructed a near-optimal column mixed strategy. Specifically, for the distribution $\tilde{\mathbf{q}}$ defined by

$$\tilde{\mathbf{q}}(j) = \frac{|\{t : j^{(t)} = j\}|}{T},$$

we showed that

$$\min_i A(i, \tilde{\mathbf{q}}) \geq \lambda^* - \epsilon.$$

Suppose instead that t is chosen uniformly at random from $[T]$. Then

$$\mathbb{E} \left[\min_i A(i, j^{(t)}) \right] \geq \lambda^* - \epsilon.$$

Multiple-Choice Questions (3 marks: 1 mark per question)

For each of the following questions, **select the correct answer** from the five choices. Each question has exactly one correct answer.

1. What is the conductance of an n -vertex path, where n is even?
 - (a) $2/n$
 - (b) $2/(n-1)$
 - (c) $1/n$
 - (d) $1/(n-1)$
 - (e) None of the above
2. What is the mixing time $T_{\text{mix}}(G, 0.01)$ of an n -vertex star graph ($n \geq 3$)?
 - (a) $\Theta(1)$
 - (b) $\Theta(\log \log n)$
 - (c) $\Theta(\log n)$
 - (d) $\Theta(n)$
 - (e) None of the above
3. In the problem of learning from expert advice, we showed in class that when there is a perfect expert, there exists a deterministic strategy that makes at most $\log n$ mistakes. Now consider a setting in which a $\Theta(1/\sqrt{\log n})$ fraction of the n experts are perfect. What is the worst-case number of mistakes made by the best possible algorithm in this setting?
 - (a) $\Theta(1)$
 - (b) $\Theta(\log \log n)$
 - (c) $\Theta(\sqrt{\log n})$
 - (d) $\Theta(\log n)$
 - (e) None of the above

Proof-Based Questions (9 marks: 3 marks per question)

1. Let $G = (V, E)$ be a graph with minimum degree δ . Suppose $(S, V \setminus S)$ is a cut with conductance

$$\Phi(S) \leq \frac{1}{2}.$$

Prove that both sides of the cut must be large, namely,

$$|S| \in \Omega(\delta) \quad \text{and} \quad |V \setminus S| \in \Omega(\delta).$$

2. Let $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ be an unknown Boolean function that is both **monotone** and **symmetric**. You may query f at any point $x \in \{-1, 1\}^n$.

Design a *deterministic* algorithm that determines f in the fewest number of queries in the worst case. Prove that your algorithm is *correct* and *optimal*.

3. Consider a variant of the **deterministic weighted majority algorithm** for learning from expert advice with n experts using **exponential updates**.

Initially, all experts have a weight of $w_i^{(1)} = 1$. At each round t , the algorithm predicts up/down according to the weighted majority vote of the experts. After observing the correct outcome, the weights are updated as follows: for every expert i that predicts incorrectly, its weight is updated by

$$w_i^{(t+1)} = w_i^{(t)} \cdot e^{-\eta},$$

where $\eta > 0$ is a fixed parameter, while the weights of correct experts remain unchanged.

Let M denote the total number of mistakes made by the algorithm, and let M_i denote the number of mistakes made by expert i . Prove that

$$M \leq \frac{M_i \eta + \ln n}{\ln\left(\frac{2}{1+e^{-\eta}}\right)}.$$

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Answer Sheet

(Do not write your name)

Student Number: _____

True or false	Answer
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	

Multiple choice questions	Answer
1	
2	
3	

For the proof-based questions, write your solutions in any available space on the answer sheet.

(Clearly indicate the question number for each of your solution)

Questions	Marks	Remarks
True/false 1–13		
MCQ 1–3		
Proof-based questions 1–3		
Total		

Table 1: This portion is for examiner's use only.

For the proof-based questions, write your solutions in any available space on the answer sheet.

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— End of the answer sheet —